Inho Choi

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Education

National University of Singapore — Singapore

- *Ph.D. in Computer Science (Advisor: Jialin Li)* — *August 2019 – February 2026*
  GPA 4.58/5.00

Yonsei University — Seoul, Republic of Korea

- *Bachelor's Degree in Computer Science* — *March 2012 – August 2019*
  GPA 3.52/4.30

Uppsala University — Uppsala, Sweden

- *Exchange Student in Information Technology* — *August 2017 – January 2018*
  GPA 4.50/5.00 (Courses: Data Mining, AI, HCI, Compiler Design)

Publications

- **[SIGCOMM '26] Capybara: Dynamic Load Balancing with Microsecond-Scale TCP Migration**
  Inho Choi, Nimish Wadekar, Guangda Sun, Raj Joshi, Joshua Fried, Omar S. Navarro Leija, Dan R. K. Ports, Irene Zhang, and Jialin Li.
  Proceedings of the 2026 ACM SIGCOMM Conference (SIGCOMM '26)
- **[APSYS '25] ML-native Dataplane Operating Systems**
  Inho Choi, Anand Bonde, Jing Liu, Joshua Fried, Irene Zhang, Jialin Li.
  16th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys 2025)
- **[ArXiv 2023] A Primer on RecoNIC: RDMA-enabled Compute Offloading on SmartNIC**
  Guanwen Zhong, Aditya Kolekar, Burin Amornpaisannon, **Inho Choi**, Haris Javaid, Mario Baldi.
  ArXiv, Dec, 2023
- **[APSYS '23] Capybara: μSecond-scale live TCP migration**
  Inho Choi, Nimish Wadekar, Raj Joshi, Joshua Fried, Dan R. K. Ports, Irene Zhang, and Jialin Li.
  14th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys 2023)
- **[SIGCOMM '23] Network Load Balancing with In-network Reordering Support for RDMA**
  Cha Hwan Song, Xin Zhe Khooi, Raj Joshi, **Inho Choi**, Jialin Li, and Mun Choon Chan.
  Proceedings of the 2023 ACM SIGCOMM Conference (SIGCOMM '23)
- **[NSDI '23] Hydra: Serialization-Free Network Ordering for Strongly Consistent Distributed Applications**
  Inho Choi, Ellis Michael, Yunfan Li, Dan R. K. Ports, and Jialin Li.
  Proceedings of the 20th USENIX Conference on Network Systems Design and Implementation
- **[S&P '20] A Stealthier Partitioning Attack against Bitcoin Peer-to-Peer Network**
  Muoi Tran, **Inho Choi**, Gi Jun Moon, Anh V. Vu, and Min Suk Kang.
  In Proceedings of IEEE Symposium on Security and Privacy, May 2020

Experiences

- **National University of Singapore - Postdoctoral Research Fellow** — Singapore
  School of Computing (Advisor: Jialin Li) — *February 2026 — Present*

- **Microsoft Research - PhD Research Intern** Redmond, WA, USA
  Systems Research Group (Mentor: Irene Zhang) *June 2025 — September 2025*

  Conducted research on ML-native dataplane OS architecture for automatic parameter tuning to address performance optimization challenges in modern datacenters at microsecond scales. Designed system architecture integrating machine learning natively into the OS dataplane for dynamic parameter optimization. Developed and evaluated the approach to demonstrate benefits of automated tuning across diverse workloads. Published initial findings at APSys '25, with full-system development currently in progress.

- **Microsoft Research - PhD Research Intern** Redmond, WA, USA
  Systems Research Group (Mentor: Dan R. K. Ports) *June 2024 — August 2024*

  Conducted research on dynamic Layer-4 load balancing to address tail latency issues under unpredictable workloads and traffic bursts due to static TCP connection assignment between client and server. Designed and implemented a new dynamic load balancing solution with microsecond-scale TCP connection migration, leveraging programmable switches and kernel-bypass technologies. Published initial findings at APSys '23, with full paper currently under submission.

- **AMD - PhD Research Intern** Singapore
  Xilinx Lab - FPGA / System Design (Mentor: Guanwen Zhong) *May 2023 - August 2023*

  Contributed to research on hardware accelerator architectures for datacenter networks, specializing in RDMA protocol optimization and computation offloading on FPGA-based SmartNIC platforms. Conducted end-to-end validation of the RecoNIC platform and implemented a representative RDMA-based hardware acceleration use case (systolic-array matrix multiplication), demonstrating direct device-memory data placement and FPGA-accelerated computation, with findings published on ArXiv.

- **National University of Singapore - Research Intern** Singapore
  Systems & Network Security Lab (Advisor: Min Suk Kang) *October 2018 - February 2019*

  Contributed to research on network-level security vulnerabilities in Bitcoin's peer-to-peer network. Participated in demonstrating partitioning attacks on Bitcoin's peer-to-peer network and defense mechanisms. Co-authored a paper published at IEEE S&P '20, advancing cryptocurrency network security.

- **Metlife - Summer Intern** Seoul, Korea
  IT Planning Team *July 2018 - August 2018*

  Analyzed IT infrastructure and database server architecture at MetLife Korea, investigating OAuth 2.0 protocol adoption for security enhancement. Presented findings through internal seminar.

Awards

- **Research Incentive Award** October 2023
  National University of Singapore
- **NUS Research Scholarship** August 2019 – July 2023
  National University of Singapore
- **Research Achievement Award** January 2023
  National University of Singapore
- **Honors - 1st Semester, 2018** August 2018
  Yonsei University

Teaching Experiences

- **Teaching Assistant for Distributed Systems (CS5223)** January – May 2022
  National University of Singapore
- **Teaching Assistant for Introduction to Operating Systems (CS2106)** August – December 2021
  National University of Singapore

- **Teaching Assistant for Distributed Systems (CS5223)** — January – May 2021
  National University of Singapore
- **Teaching Assistant for Programming Methodology (CS1010E)** — August – December 2020
  National University of Singapore
- **Teaching Assistant for Cryptography Theory and Practice (CS4236)** — August – December 2019
  National University of Singapore

Mentoring Experiences

- **Mohsen Ghasemi — Undergraduate Student, NUS** — 2025
  Kernel-Bypass Network Stack Optimization
- **Junhao Yang — Undergraduate Student, NUS** — 2025
  Kernel-Bypass Network Stack Optimization
- **Nimish Wadekar — Research Assistant, NUS** — 2024 – 2025
  Implementing Microsecond-Scale Live TCP Migration in a Kernel-Bypassing Library OS
- **Yiyang Liu — Master Student, NUS** — 2024
  Thesis: Enhancing Distributed Systems with Hydra: A Software Solution for Scalable Network Ordering

Services

- **Reviewer** — ToN '25
- **Shadow PC Reviewer** — EuroSys '26, EuroSys '25
- **Student Volunteer** — SOSP '21

Talks

- **Co-Designing Systems for Microsecond-Scale Consistent Tail Latency in Datacenters**
  Yonsei University (2025), Seoul National University (2025)